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#include <ESP8266WiFi.h>

const char* ssid = "your_SSID";
const char* password = "your_PASSWORD";
WiFiServer server(80); // Set up server on port 80
// Define the GPIO pin where the LED/light is connected
const int lightPin = 2; // GPIO 2 on ESP8266
void setup() {
    Serial.begin(115200);
    pinMode(lightPin, OUTPUT);
    digitalWrite(lightPin, LOW); // Start with light off

    // Connect to Wi-Fi
    WiFi.begin(ssid, password);
    Serial.print("Connecting to WiFi");
    while (WiFi.status() != WL_CONNECTED) {
        delay(500);
        Serial.print(".");
    }
    Serial.println("\nConnected to WiFi!");
    Serial.print("IP Address: ");
    Serial.println(WiFi.localIP());

    // Start the server
    server.begin();
}

void loop() {
    WiFiClient client = server.available();
    if (!client) {
        return;
    }

    Serial.println("Client connected");
    String request = client.readStringUntil('\r');
    Serial.println(request);
    client.flush();

    // Check the request for light control
    if (request.indexOf("/light/on") != -1) {
        digitalWrite(lightPin, HIGH); // Turn the light on
    } else if (request.indexOf("/light/off") != -1) {
        digitalWrite(lightPin, LOW); // Turn the light off
    }

    // Create a simple HTML page to control the light
    client.println("HTTP/1.1 200 OK");
    client.println("Content-Type: text/html");
    client.println();
    client.println("<!DOCTYPE HTML>");
}

```

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    client.println("<html>");
    client.println("<h1>ESP8266 Light Control</h1>");
    client.println("<p><a href=\"/light/on\"><button>Turn Light  
ON</button></a></p>");
    client.println("<p><a href=\"/light/off\"><button>Turn Light  
OFF</button></a></p>");
    client.println("</html>");

    delay(1);
    Serial.println("Client disconnected");
}
```